

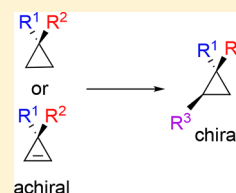
Asymmetric Preparation of Polysubstituted Cyclopropanes Based on Direct Functionalization of Achiral Three-Membered Carbocycles

Focus Review

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ABSTRACT: In addition to the appealing synthetic transformations that cyclopropanes present, they are also part of larger molecular structures that possess a wide range of biological properties. Therefore, the preparation of enantiomerically enriched cyclopropanes has consistently been a very interesting research topic in organic synthesis. In this Focus Review, we are presenting new methods for the synthesis of these target compounds through catalytic and asymmetric direct functionalization of simple achiral three-membered carbocycle precursors. These convergent and very flexible approaches allow the preparation of a large variety of polysubstituted alkyl-, vinyl-, alkynyl-, and arylcyclopropanes but also cyclopropanols and cyclopropylamines in very high diastereo- and enantiomeric ratios from a single precursor, underlining the power of this synthetic route.



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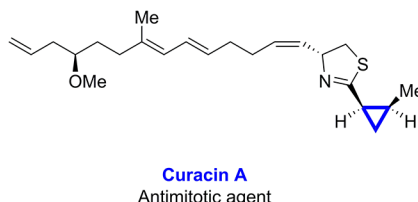
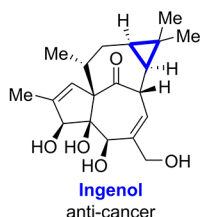
1. INTRODUCTION

The cyclopropane ring is unique among all carbocycles due to its unusual bonding and inherent ring strain and has constantly been of great scientific interest to the organic community.^{1–4} In addition to the appealing synthetic transformations that cyclopropanes might present, these smallest carbocycles are part of larger molecular structures that possess a wide range of biological properties spanning from enzyme inhibitions (MAO, DOPA, ALDH among others), antibacterial, antimetastatic (anticancer), antifungal, insecticidal, plant growth, and fruit ripening control (Figure 1).^{5–14} There are several chemical mechanisms through which they achieve their biological activities such as addition to the cyclopropane bond, C–H

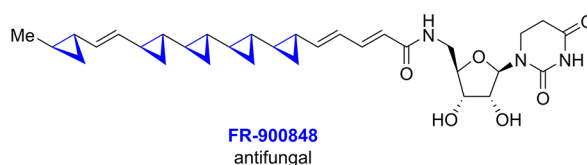
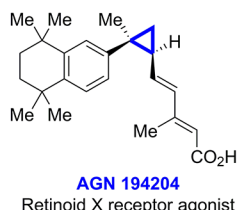
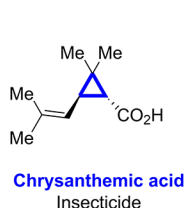
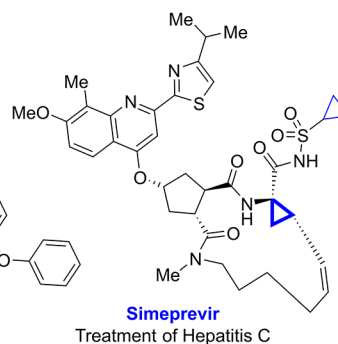
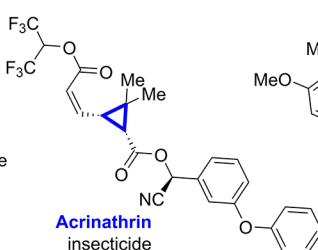
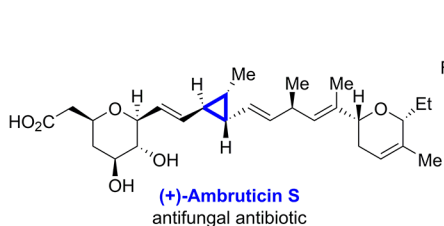
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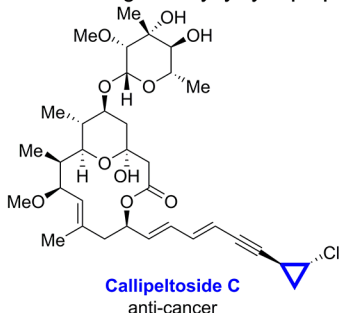
a) Molecules containing alkylcyclopropane structures



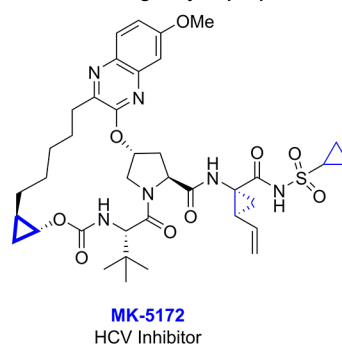
b) Molecules containing vinylcyclopropane structures



c) Molecule containing an alkynylcyclopropane structure



d) Molecule containing a cyclopropanol structure



e) Molecule containing cyclopropylamine structures

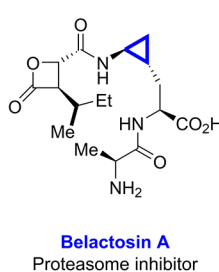
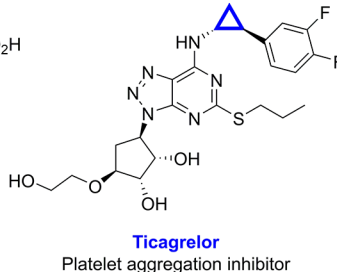
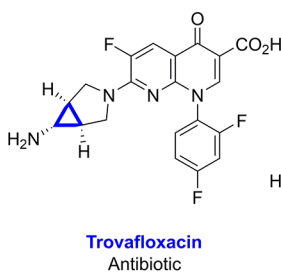
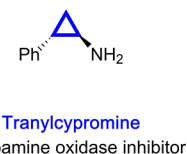
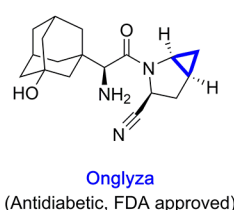
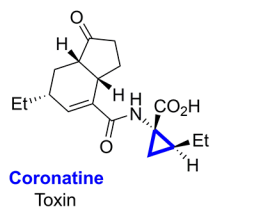
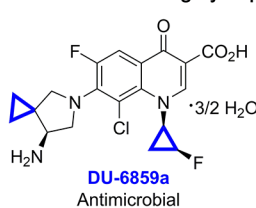
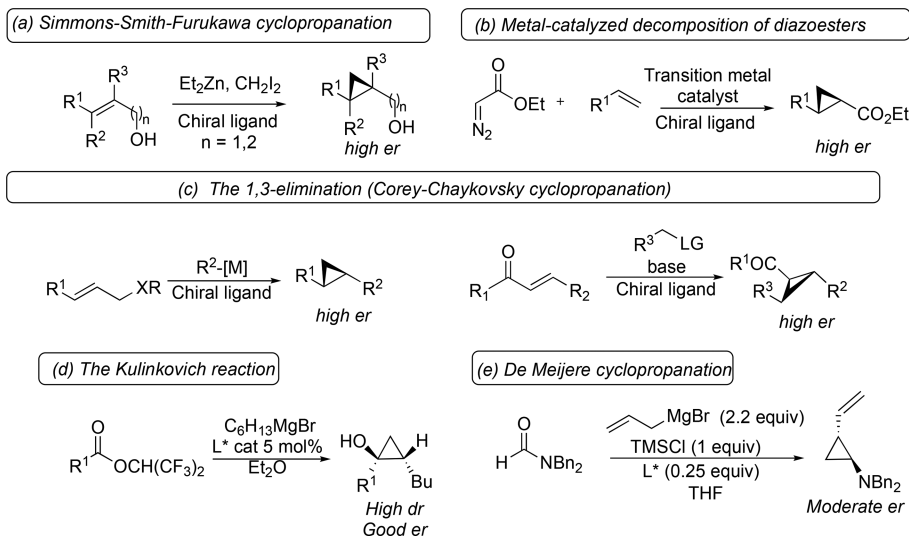


Figure 1. Selected examples of natural products and biological active molecules containing the cyclopropyl core.

oxidation, radical chemistry, two electrons oxidation, nucleophilic substitution, and electrophilic ring opening.¹⁵ From a more medicinal chemistry perspective, the cyclopropane ring is

also able to adopt unique conformations allowing interesting spatial orientation of the substituents of the cyclopropyl subunit. A prime example is the case of MK-5172 developed by Merck

Scheme 1. Main Methods for the Asymmetric Preparation of Functionalized Polysubstituted Cyclopropanes



which is currently in clinical trials for the treatment of hepatitis C.¹⁶ Placement of the cyclopropyl ring on the macrolactone allowed for certain conformational constraints enhancing the inhibitor potency. Additionally, the cyclopropyl ring also serves as a bioisostere for lipophilic alkyl chains.¹⁷ It should be emphasized that all types of cyclopropyl rings could be found as active compounds spanning from classical alkyl- (Figure 1a), to vinyl- (Figure 1b), to alkynyl- (Figure 1c) cyclopropanes but also to α -heterosubstituted cyclopropanes possessing either an alkoxy (cyclopropanol derivatives, Figure 1d) or an amino (cyclopropylamine derivatives, Figure 1e) group as described in Figure 1.

In addition to their biological activities and as a logical consequence from their less than ideal 60° bond angles, the considerable torsional and angular strains make them valuable synthetic building blocks.^{18–20} For instance, cyclopropanes can undergo various chemical transformations such as ring expansion,^{21–24} selective ring opening,^{25–31} carbon–carbon cleavage,^{32–42} cycloaddition,^{43–46} and rearrangement.^{47–50} Donor–acceptor cyclopropanes have also participated in various transformations^{51–53} and were applied to the synthesis of a potent natural antibiotic ($\pm$)-platensimycin, that showed outstanding activity against methicillin-resistant *Staphylococcus aureus* (MRSA).^{54–57} From a rapid examination of all the structures summarized in Scheme 1, the preparation of a variously substituted cyclopropyl ring is an already well-investigated topic of research, abundantly reviewed over the years.^{58–60} The most classical methods for the asymmetric preparation of cyclopropane derivatives could be classified into the following four main categories: (a) The Simmons–Smith–Furukawa cyclopropanation reaction (Scheme 1a); (b) The metal-catalyzed decomposition of diazoester (Scheme 1b); (c) The sequence Michael addition–1,3-elimination reaction also called the Michael initiated ring closures reaction (MIRC) such as the Corey–Chaykovsky cyclopropanation reaction (Scheme 1c), the Kulinkovitch reaction for the preparation of cyclopropanols (Scheme 1d), and the De Meijere variant of the latter approach for the synthesis of cyclopropylamine derivatives (Scheme 1e).

Charette's enantioselective Simmons–Smith–Furukawa cyclopropanation reaction, relying on the coordination of the incoming organozinc reagent to a Lewis base, is a very well

developed reaction with a broad scope encompassing nowadays the initial allylic alcohols (Scheme 1a).^{61–63} The cyclopropanation reaction of olefins using transition-metal-catalyzed decomposition of diazoalkanes is also an extremely well-investigated transformation for the synthesis of cyclopropyl esters (Scheme 1b).^{64,65} Both inter- and intramolecular versions of these reactions were developed and nicely exploited in synthesis. The most exhaustively studied diazo reagents for intermolecular cyclopropanation reactions are diazoesters or azo-compounds containing other electron-withdrawing groups. Although suitable procedures are available for their synthesis, diazoalkanes other than diazomethane are more challenging to prepare, inherently unstable, and could be explosive in pure form.^{66,67} A wide range of metal catalysts derived from Cu, Rh, Ru, Co, Fe, Os, Pd, Pt, Cr, and others have been reported to catalyze the decomposition of the diazo reagent for which an incredibly large number of chiral complexes have been prepared and tested. Today, this method is widely used to generate cyclopropanes substituted with electron withdrawing groups including the formation of cyclopropanols and cyclopropylamines through the addition of diazoester to α -heterosubstituted alkenes ($R^1 = \text{OR}, \text{NR}_2, \text{N}_3$). The generation of a nucleophile which undergoes an addition followed by an elimination reaction, coined the Michael initiated ring closures reaction (MIRC), has been abundantly used in synthesis and only two representative examples are discussed therein. The Corey–Chaykovsky cyclopropanation reaction involves the addition of a sulfur ylide to an enone, generating a transient enolate, which undergoes a subsequent intramolecular cyclization to give the cyclopropane motif.^{68–70} The second approach relies on the (–)-sparteine-catalyzed asymmetric carbolithiation of cinnamyl derivatives⁷¹ leading, after the 1,3-elimination reaction, to the corresponding *trans*-cyclopropanes in excellent enantiomeric ratios.⁷² As far as the cyclopropanols are concerned, and although cyclopropanation reactions of enol ethers using the Fischer carbene have been reported,⁷³ the most widely used method relies on the addition of a Grignard reagent to an ester in the presence of a catalytic amount of $\text{Ti}(\text{O}i\text{Pr})_4$ (Kulinkovich reaction).^{74–76} The introduction of a chiral titanium species lead to the formation of chiral cyclopropanols in good enantioselectivity (see Scheme 1d). The preparation of cyclopropylamines is based on the same conceptual approach

from various amides (Scheme 1e).^{77,78} The scope of this reaction was later expanded to the use of alkyl and aryl nitriles as suitable candidates in forging the cyclopropylamine motif.⁷⁹

However, and despite the increasing and justified popularity of these well-known methods, this short summary underlined one intrinsic problem: for every cyclopropane that one needs to prepare, a different strategy is needed limiting the rapid structural diversification usually required for biological studies. It has become clear that a complementary approach that would produce all types of polysubstituted cyclopropanes (alkylcyclopropanes, vinylcyclopropanes, arylcyclopropanes, alkynylcyclopropanes, cyclopropanols, and cyclopropylamines), through a truly catalytic asymmetric reaction and from a single and unique precursor, would certainly be highly desirable and would represent a powerful addition to the field of small ring

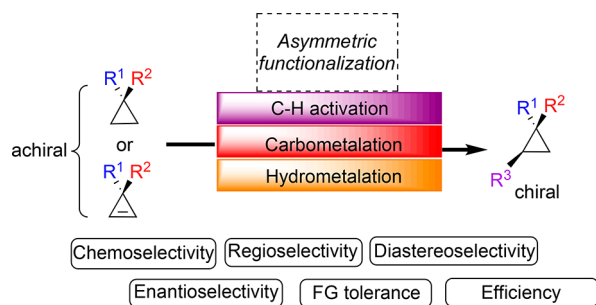


Figure 2. Asymmetric direct functionalization of achiral three-membered carbocycles.

chemistry.^{80,81} This strategy could be achieved only if one has a toolbox filled with diastereo- and enantioselective approaches to functionalize, at least, one of the carbons of an achiral three-membered ring (any chemical transformation that is not proceeding at one of the three-carbon atoms of the cyclopropyl ring will not be addressed in this Focus Review, Figure 2).

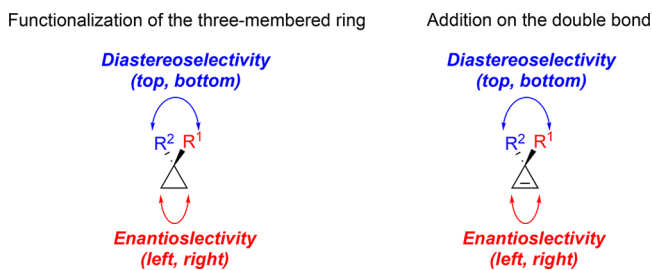


Figure 3. Diastereoselectivity and enantioselectivity in the direct functionalization of achiral three-membered carbocycles.

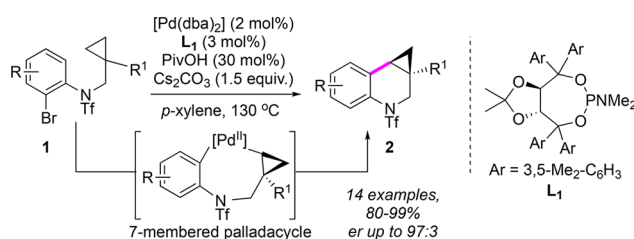
To assemble such diverse families of three-membered rings (alkylcyclopropanes, vinylcyclopropanes, arylcyclopropanes, alkynylcyclopropanes, cyclopropanols, and cyclopropylamines) in a single-pot reaction from a single and unique precursor requires the chiral catalyst to perform a double facial selection: (i) enantiotopic face selection (left or right) and (ii) diastereotopic face selection (top or bottom) as described in Figure 3.

In this review, we will describe the asymmetric functionalization of achiral cyclopropanes/cyclopropenes by manipulation of, at least, one carbon of the three-membered ring.

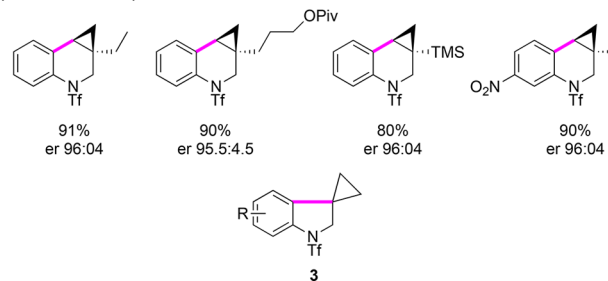
2. DIRECT FUNCTIONALIZATION OF THREE-MEMBERED CARBOCYCLES BY ASYMMETRIC C–H BOND ACTIVATION

In the past few decades, C–H bond activation has become one of the most intensively investigated research fields resulting in more efficient preparation of complex molecular architectures.^{82–91} In this regard, one challenging issue to address is the enantioselective C–H bond activation for the construction of new C–C and C–X bonds.^{92,93} However, when the metal-catalyzed direct C–H functionalization was performed on the cyclopropyl skeleton, ring fragmentations have been observed.^{25–42,47–50} Nevertheless, by a carefully designed cyclopropyl ring bearing a suitable directing group, enantioselective C–H bond activation has been reported, either intra- or intermolecularly, leading ultimately to the creation of new chiral centers.

Scheme 2. Pd-Catalyzed Enantioselective C–H Arylation of Cyclopropanes



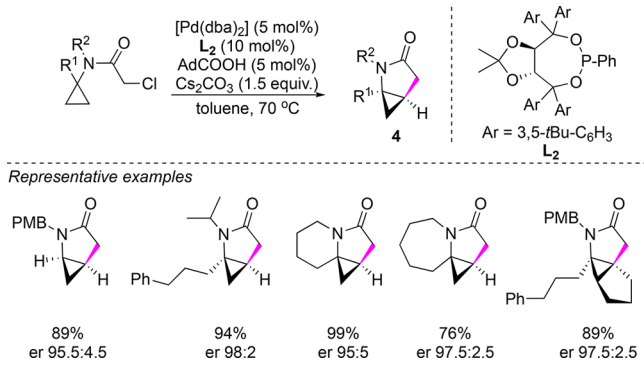
Representative examples



2.1. Intramolecular Direct Functionalization of Achiral Three-Membered Carbocycles by Asymmetric C–H Bond Activation

An initial report by Cramer and co-workers describes the intramolecular Pd/TADDOL-catalyzed asymmetric C–H bond arylation reaction of **1** to produce functionalized tetrahydroquinolines **2** in good yields and excellent enantiomeric ratios (Scheme 2).⁹⁴ The reaction tolerates several functionalities for R^1 , but the presence of a carboxylic cocatalyst (30 mol %) is essential to reach good selectivity and yield. If unsubstituted cyclopropane **1** is used ($R^1 = H$), the C–H bond activation proceeds exclusively to give spirocycle **3** indicating that a six-membered palladacycle is preferentially formed in the absence of substituent. In contrast, a nonconventional seven-membered palladacycle results when $R^1 \neq H$.

A few years after, the same group further expended this Pd-catalyzed asymmetric intramolecular C–H bond activation to the construction of azabicyclo[3.1.0]hexane scaffolds **4** (Scheme 3) in high yields and excellent enantioselectivities by using a more bulky TADDOL phosphonate ligand (L_2).⁹⁵ It is worth mentioning that when the substrate contained several active

Scheme 3. Enantioselective C–H Functionalization for the Synthesis of Azabicyclo[3.1.0]hexane

sites, the enantioselective direct functionalization occurred selectively on the cyclopropyl ring to produce a complex tricyclic skeleton under mild reaction conditions.

The combination of these last two approaches provided a direct access to dihydroquinolones and dihydroisoquinolones in excellent enantioselectivities,⁹⁶ and was successfully applied to the concise synthesis of a key intermediate en route to the synthesis of BMS-791325, an antiviral drug candidate used as HCV inhibitor (Scheme 4).⁹⁷

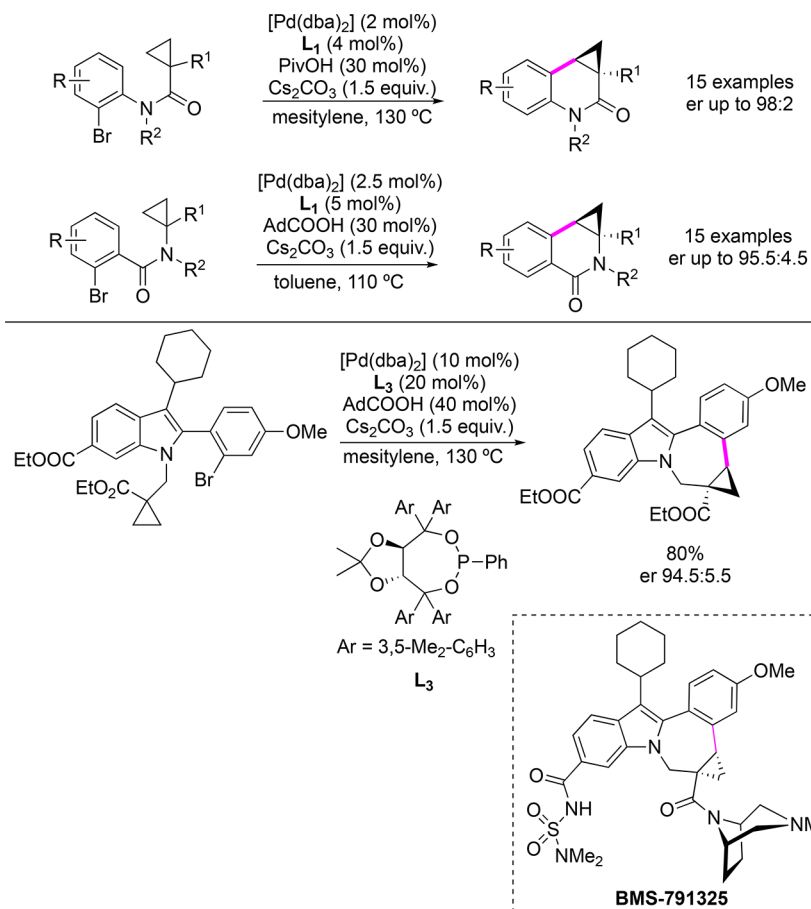
An additional intriguing subset of saturated *N*-heterocyclic scaffolds that displays an interesting potential for pharmaceutical drugs are 3-azabicyclo[3.1.0]hexanes that could also be achieved

by intramolecular asymmetric C–H bond activation. For instance, Cramer realized the catalytic and asymmetric preparation of 3-azabicyclo[3.1.0]hexenes **6** using trifluoroacetimidoyl chloride **5** as substrates for C–H bond cyclization reaction (Scheme 5).⁹⁸ Then, the electrophilic ketimine reacts diastereoselectively with various nucleophiles to provide functionalized 3-azabicyclo[3.1.0]hexanes.

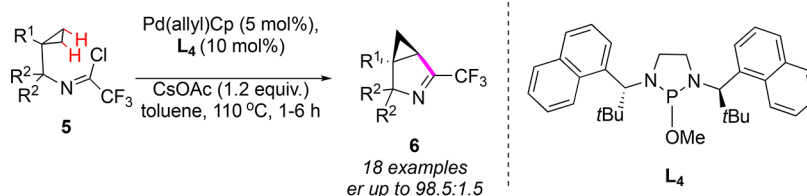
The asymmetric intramolecular C–H bond activation creating a carbon–silicon bond on the cyclopropyl ring has recently been realized by Hartwig opening the door to the preparation of heterosubstituted cyclopropanes.⁹⁹ The initial $[\text{Ru}(\text{PPh}_3)_3\text{Cl}_2]$ -catalyzed dehydrogenative coupling of cyclopropylmethanol **7** with dimethylsilane afforded the intermediate hydroxyl silyl ether that was used without any purification for a subsequent $[\{\text{Rh}(\text{cod})\text{Cl}\}_2]/(S)\text{-DTBM-SEGPHOS}$ -catalyzed transformation, in the presence of cyclohexene as hydrogen acceptor. Oxasilolanes **8**, possessing a rather large variety of functional groups on the aromatic rings ($\text{R} = \text{Ar}$), were obtained in excellent enantiomeric excesses (Scheme 6). When R is an alkyl group, lower enantioselectivities were observed. Enantioenriched oxasilolanes **8** are easily converted into cyclopropanols by Tamao–Fleming oxidation reaction.¹⁰⁰

2.2. Interolecular Direct Functionalization of Achiral Three-Membered Carbocycles by Asymmetric C–H Bond Activation

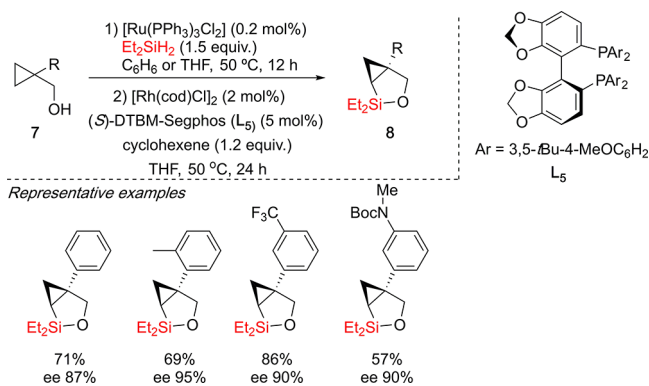
Further demonstrating the potential of this approach, the asymmetric functionalization of three-membered carbocycles by

Scheme 4. Enantioselective Pd-Catalyzed Intramolecular Cyclopropane Functionalization: Access to Dihydroquinolones, Dihydroisoquinolones, and BMS-791325 Ring System

Scheme 5. Enantioselective C–H Functionalization en Route to 3-Azabicyclo[3.1.0]hexenes



Scheme 6. Rhodium-catalyzed enantioselective silylation of cyclopropyl C–H bonds



intermolecular C–H bond activation represents also an alternative route to optically active cyclopropanes bearing different functionalities. For instance, chiral mono-*N*-protected amino acid ligand L₆ could promote the enantioselective Pd-catalyzed intermolecular C–H activation of cyclopropanes which was subsequently followed by an *in situ* cross-coupling reaction with either aryl, vinyl, or alkylboron reagents. *Cis*-substituted chiral cyclopropyl carboxylic acids **9** were obtained in good enantiomeric excesses (Scheme 7).¹⁰¹

Later, the same group extended this concept to the Pd(II)/Pd(IV)-catalyzed enantioselective C–H arylation reaction of triflyl-protected cyclopropylmethyamines **10** with a variety of aryl iodides by using the mono-*N*-protected amino acid ligand L₇.¹⁰² Optically enriched functionalized three-membered rings **11** were obtained with high enantioselectivities (Scheme 8). It is important to note that the triflyl protecting group could be removed at the end of the sequence to produce chiral cyclopropylmethyamine derivatives with identical enantiopurity.

Furthermore, by using the chiral acetyl-protected amino-methyl oxazoline ligand, achiral cyclopropane carboxylic amide underwent an asymmetric direct borylation reaction to produce optically active cyclopropylboronate in moderate 32% yield but with an excellent enantioselectivity.¹⁰³

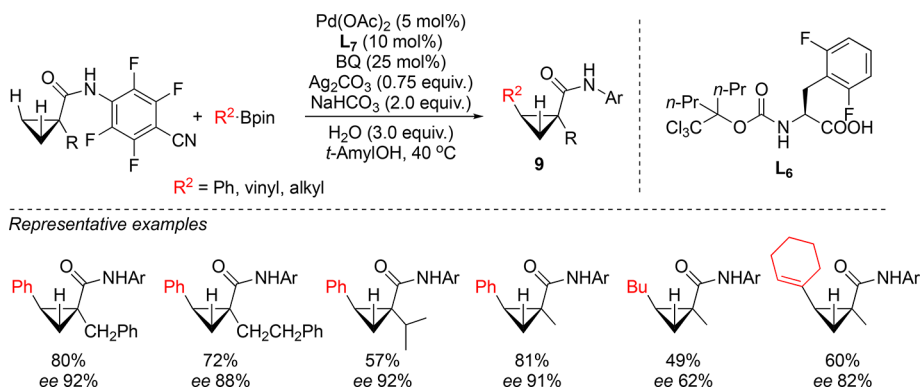
A monoprotected aminoethyl amine chiral ligand based on an ethylenediamine backbone was developed to achieve Pd-catalyzed enantioselective C(sp³)-H arylation of cyclopropane-carboxylic acid without exogenous directing groups. The majority of the aryl iodides containing electron-withdrawing and electron-donating groups afforded the desired products in good yields and high enantioselectivities. Similarly, a wide range of α -substituted cyclopropanecarboxylic acid was also tested using methyl iodobenzoate as the coupling partner and similar selectivities were observed (Scheme 9).¹⁰⁴

If one wants to add a large variety of electrophiles from a single enantioenriched intermediate, the asymmetric β -metalation of cyclopropylcarboxamide **12** still remains a powerful approach. Using the directing carboxamide substituent, a *syn*-metalation of the cyclopropyl ring smoothly occurred at low temperature in the presence of *s*-BuLi/(–)-sparteine to provide the diastereoisomerically pure and enantiomerically enriched corresponding cyclopropyllithium species **13** (Scheme 10).¹⁰⁵ Addition of a large variety of electrophiles gave the expected cyclopropanes **14** in very good enantiomeric excess.

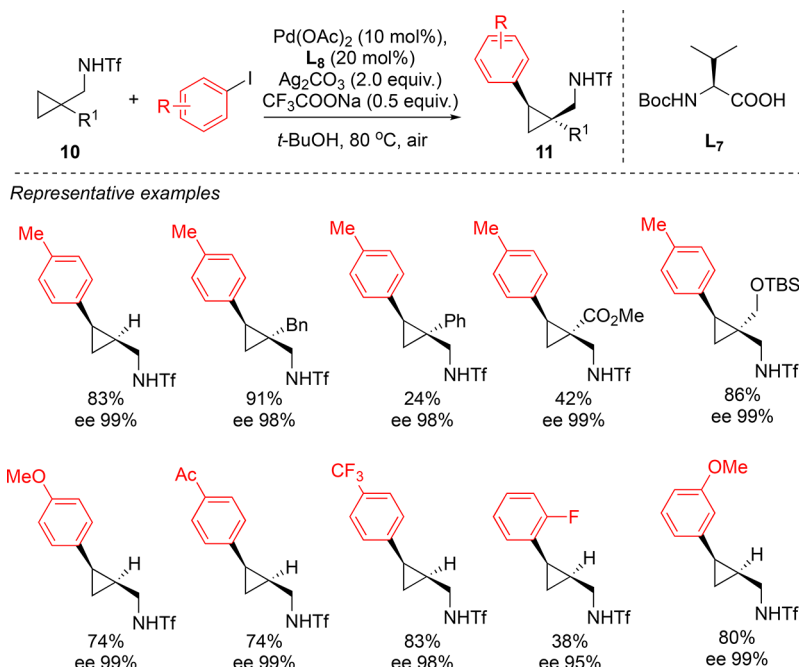
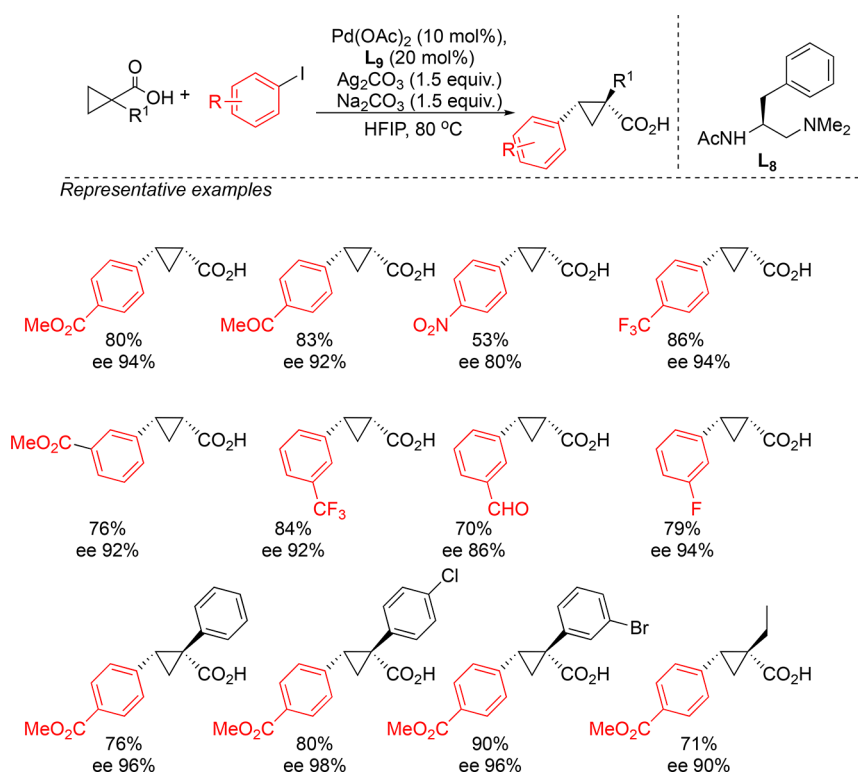
3. DIRECT FUNCTIONALIZATION OF ACHIRAL THREE-MEMBERED CARBOCYCLES BY ASYMMETRIC ADDITION ON THE DOUBLE BOND

Comparing the heat of hydrogenation for the conversion of cyclopropene to cyclopropane with the conversion of acetylene to ethylene (54 versus 42 kcal·mol^{–1}, respectively) indicates that cyclopropenes have similar reactivity than alkynes,¹⁰⁶ that was also supported by numerous theoretical investigations.^{107–111} This simplified comparison suggests that reactions classically performed on alkynes such as hydrometalation and carbometalation reactions should also be possible with cyclopropenes.

Scheme 7. Pd(II)-Catalyzed Enantioselective C–H Activation of Cyclopropanes



Scheme 8. Pd(II)-Catalyzed Highly Enantioselective C–H Arylation of Cyclopropylmethylamines

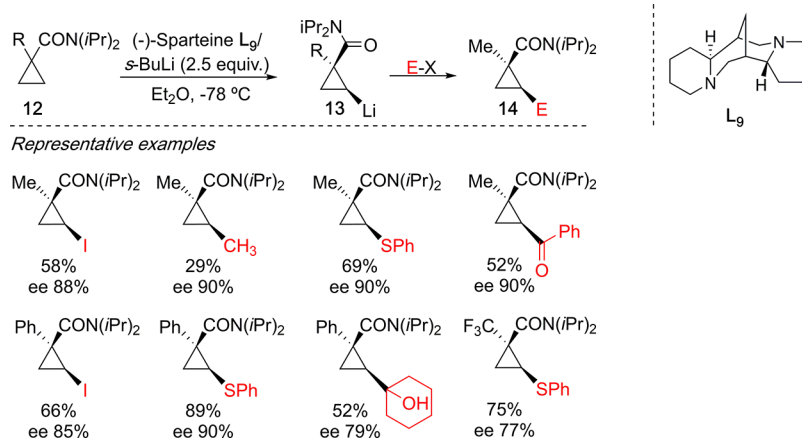
Scheme 9. Pd(II)-Catalyzed Enantioselective C(sp³)-H Arylation of Free Carboxylic Acids

3.1. Direct Functionalization of Achiral Three-Membered Carbocycles by Asymmetric Hydrogenation Reactions

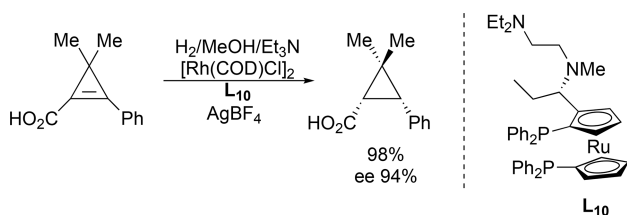
A single example of the highly efficient catalytic enantioselective hydrogenation of achiral cyclopropane was reported. This transformation was catalyzed by a Rh(I) complex bearing a ruthenocene-based chiral diphosphine ligand L_{10} and was unfortunately limited to tetrasubstituted cyclopropenes bearing a carboxylic group at the double bond (Scheme 11).¹¹²

3.2. Direct Functionalization of Achiral Three-Membered Carbocycles by Asymmetric Hydroboration Reactions

The pioneering Rh-catalyzed asymmetric hydroboration reactions of cyclopropenes **15** possessing an ester moiety were reported by Gevorgyan in 2003 to produce the enantioenriched cyclopropyl boronates **16** with an excellent degree of diastereoselectivity and enantioselectivity (Scheme 12).¹¹³ Similar to the ester moiety, a methoxymethyl substituent serves also as a good directing group leading virtually to a single

Scheme 10. Enantioselective Metalation of Cyclopropylcarboxamides Using *s*-BuLi-Sparteine Complex

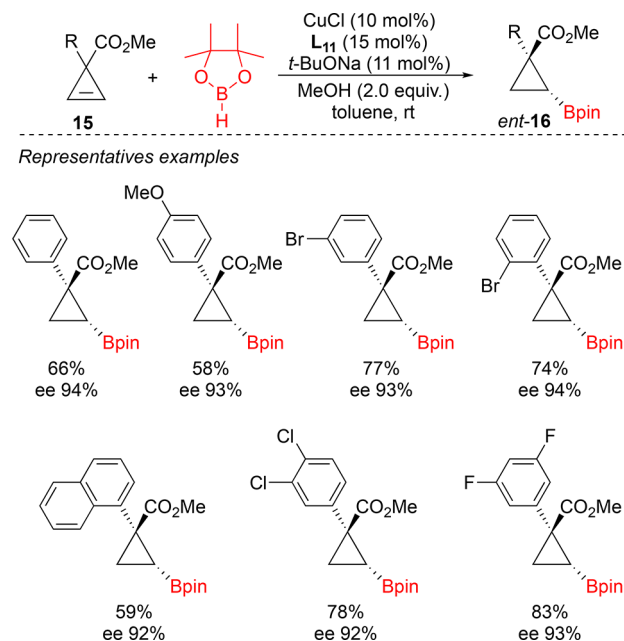
Scheme 11. Rh(I)-Catalyzed Asymmetric Hydrogenation of Fully-Substituted Cyclopropenes



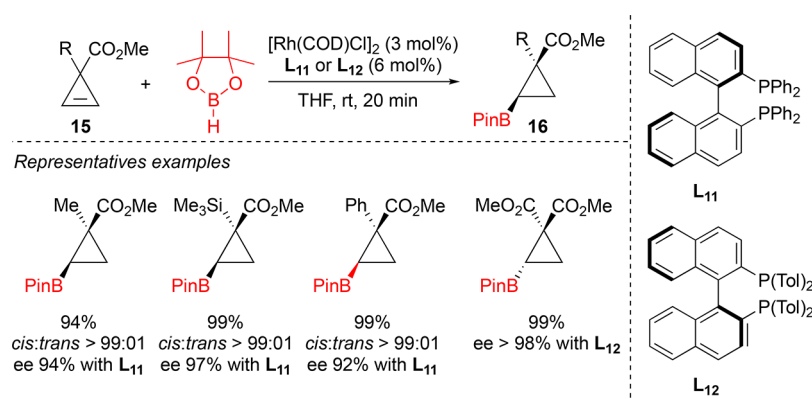
diastereoisomer with high enantioselectivity. To further demonstrate the synthetic potential of these products, Suzuki cross-coupling reactions with aryl and vinyl iodide derivatives were tested. Although cyclopropylboronic esters were reluctant to react, the corresponding cyclopropylboronic acids provided good yield of the coupled products with pure retention of configuration under Fu's reaction condition.¹¹⁴ In 2017, a detailed study on the Rh-catalyzed asymmetric hydroboration of achiral cyclopropenes containing an ester or amide directing group was reported.¹¹⁵

The opposite diastereoisomer (Bpin *trans* to the ester) was obtained with a different catalytic system composed of CuCl with (*R*)-BINAP L₁₁ under basic conditions (Scheme 13).¹¹⁶ Labeling experiments show that the reaction proceeds though the classical *syn* addition but surprisingly from the opposite face, *syn* to the aryl group.

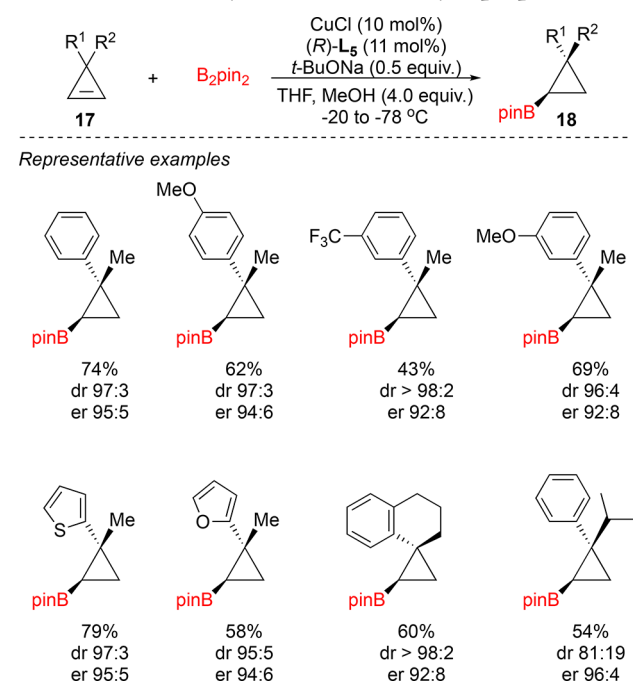
Scheme 13. Copper-Catalyzed Enantioselective Hydroboration of Cyclopropenes



Scheme 12. Rh-Catalyzed Enantioselective Hydroboration of Cyclopropenes



Scheme 14. Copper-Catalyzed Diastereo- and Enantioselective Desymmetrization of Cyclopropenes



In the two previous cases (Schemes 12 and 13), the presence of an ester (or methoxymethyl group) was essential to reach high enantioselectivity (Scheme 12) or even reactivity (Scheme 13). However, Tortosa subsequently developed a catalytic system that was successful for nonfunctionalized cyclopropenes 17.¹¹⁷ Her approach consisted in the addition of B_2Pin_2 combined with a catalytic amount of $CuCl$ and (*R*)-DTBM-Segphos under mild basic conditions (Scheme 14). Cyclopropenes possessing electron rich as well as electron poor aromatic rings led similarly to polysubstituted cyclopropylboronic esters 18 with excellent enantiomeric excesses.

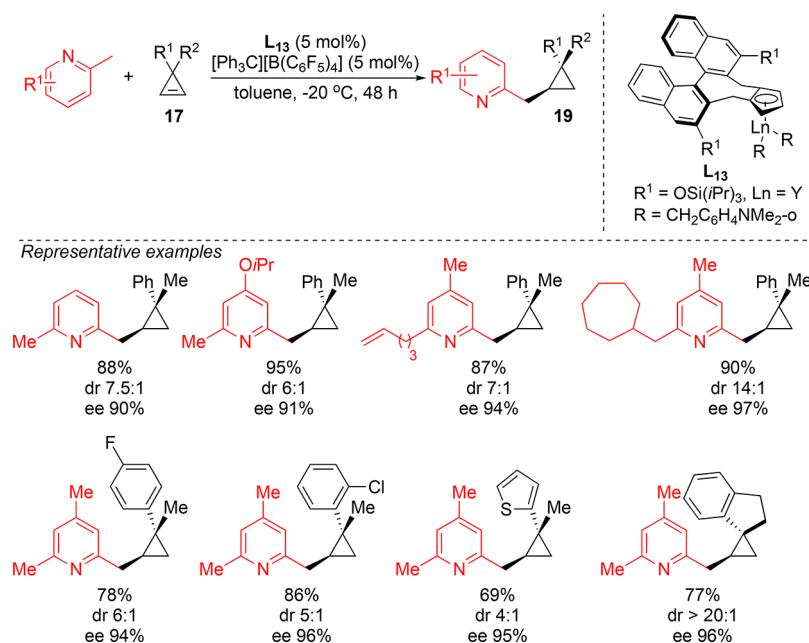
3.3. Direct Functionalization of Achiral Three-Membered Carbocycles by Asymmetric Hydroalkylation Reactions

A unique asymmetric yttrium-catalyzed addition of 2-methyl-azaarenes to cyclopropenes 17 provides access to pyridylmethyl-functionalized cyclopropanes 18 in high yields and diastereo- and enantioselectivities (Scheme 15).¹¹⁸ This study underlined the propensity of half-sandwich rare-earth metal complexes to serve as an efficient catalyst for the addition of a large variety of heteroaromatic compounds such as pyridine, anisoles, and *N,N*-dimethylanilines to the strained double bond of cyclopropenes.

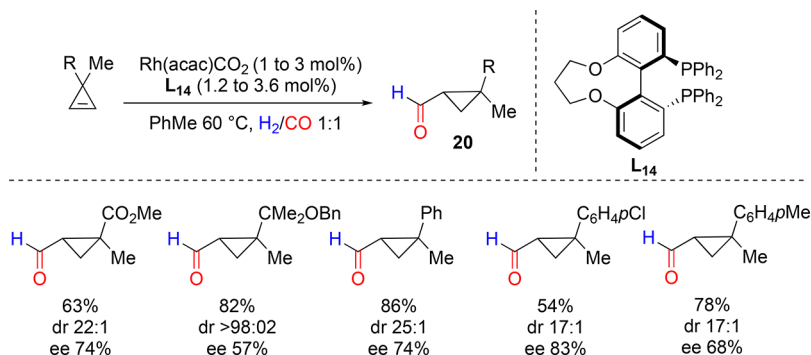
3.4. Direct Functionalization of Achiral Three-Membered Carbocycles by Asymmetric Hydroformylation Reactions

Rubin and co-workers have reported the stereoselective (*R*)-C3-Tunephos/Rh-catalyzed hydroformylation of cyclopropenes to produce the desired cyclopropylcarboxaldehydes 20 with very good diastereo- but moderate enantioselectivity (Scheme 16).¹¹⁹ This unique combination avoids the formation of the migratory insertion of the metal into the C–C bond of the cyclopropenyl ring as well as the often encountered [2 + 2] dimerization of cyclopropenes.

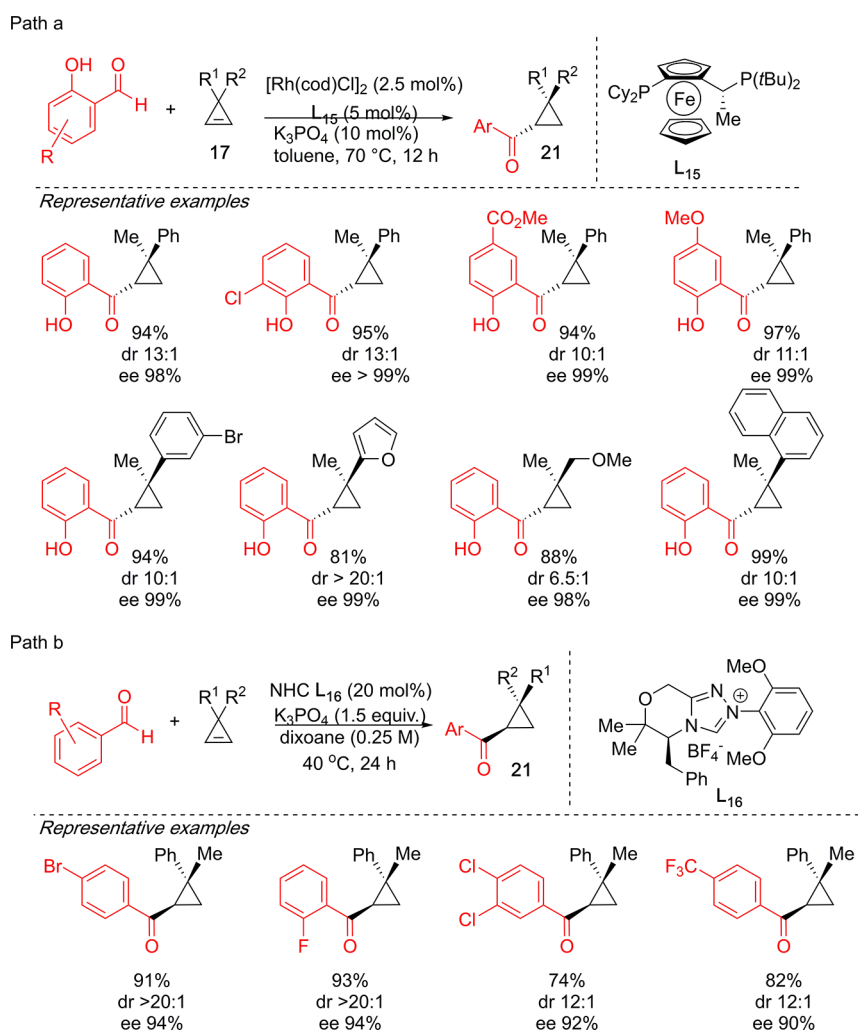
To prepare cyclopropylketone derivatives, an alternative approach would consist in the metal-catalyzed hydroacylation reaction. Potential decarbonylation has already occurred in classical hydroacylation reactions, but the strain energy released in the proposed reaction should drive the reaction to completion. In this context, Dong's research group could perform the desymmetrization of cyclopropenes 17 by an intermolecular Josiphos L_{15} /Rh-catalyzed hydroacylation reaction with salicylaldehyde, known to coordinate the Rh and promote the reaction.¹²⁰ Various aryl cyclopropylketones 21 containing quaternary carbon stereocenters were therefore obtained with high diastereoselectivity and enantioselectivity (Path a, Scheme 17). Chiral electron rich *N*-heterocyclic carbenes L_{16} (NHCs) could react with aldehydes to produce the Breslow intermediate that subsequently reacts with cyclopropenes to give the same enantiomerically enriched cyclopropylketones 21 (Path b, Scheme 17).¹²¹

Scheme 15. Asymmetric Yttrium-Catalyzed C(sp³)-H Addition of 2-Methyl Azaarenes to Cyclopropenes

Scheme 16. Rhodium-Catalyzed Hydroformylation of Cyclopropenes



Scheme 17. Rh-Catalyzed and N-Heterocyclic Carbene Catalyzed Acylation Reaction



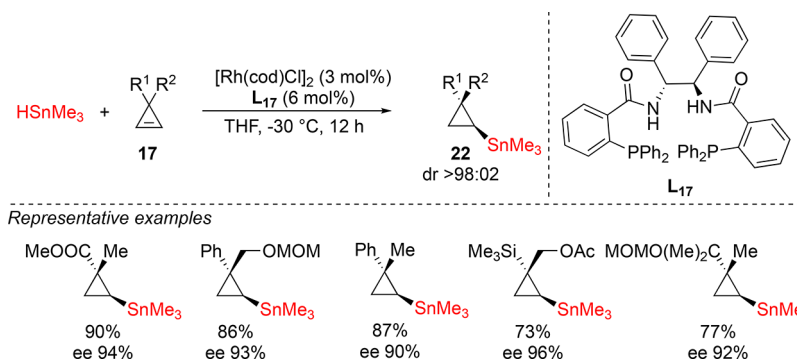
3.5. Direct Functionalization of Achiral Three-Membered Carbocycles by Asymmetric Hydrostannation Reactions

In the presence of chiral rhodium catalyst, Gevorgyan and co-workers have reported the asymmetric hydrostannation of cyclopropenes **17** to produce enantioenriched cyclopropylstannane **22** with high diastereo- and enantioselectivity.¹²² It should be noted that replacement of Me_3SnH by Bu_3SnH leads surprisingly to racemic products whereas a more substituted cyclopropene did not lead to the hydrostannation reaction (Scheme 18).

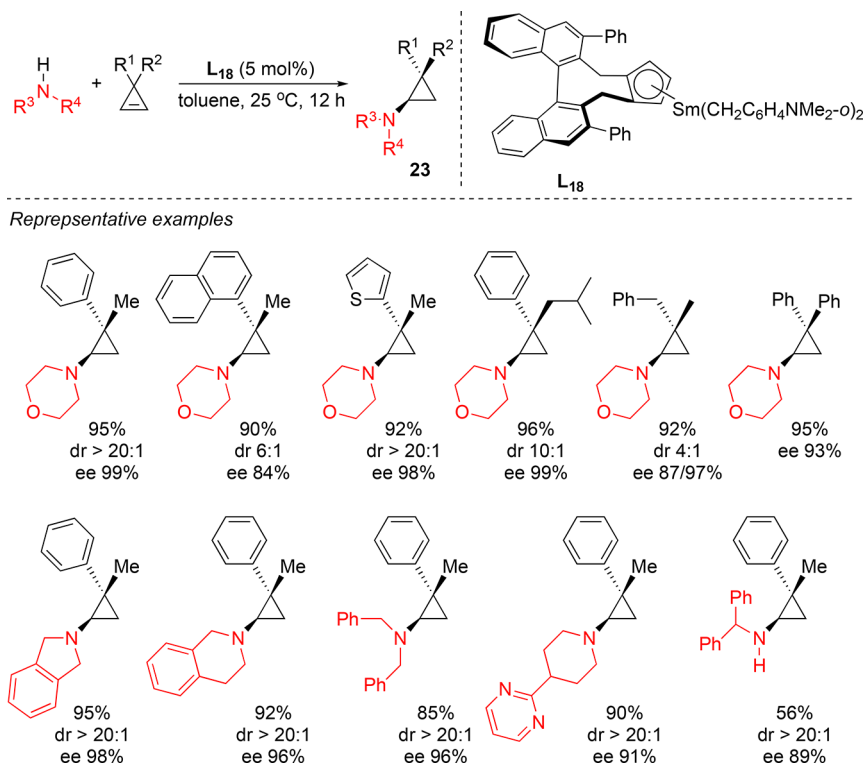
3.6. Direct Functionalization of Achiral Three-Membered Carbocycles by Asymmetric Hydroamination Reactions

As chiral half-sandwich rare-earth-metal complex L_{11} acted as efficient catalyst for the addition of 2-methylazaarenes to cyclopropenes, the asymmetric addition of amines was then successfully developed. Chiral cyclopropylamines **23** were obtained when the Sm complex L_{18} was added to secondary amines (Scheme 19).¹²³ Various cyclopropenes as well as amines were tested and excellent diastereoselectivities and enantioselectivities were obtained in all cases.

Scheme 18. Rh-Catalyzed Hydrostannation of Cyclopropenes



Scheme 19. Sm-Catalyzed Cyclopropenes Hydroamination



If the amines possess an electrophilic site, the use of a catalyst half-sandwich lanthanum complexes **L**₁₉ allows the diastereo- and enantioselective addition of allyl and propargyl amines on the strained double bond of the cyclopropenes followed by a fast intramolecular carbometalation reaction on the pendant unsaturated moiety.¹²⁴ Bicyclic aminocyclopropanes **24** were obtained with very high selectivities (Path a, Scheme 20). When the half-sandwich lanthanum complex **L**₂₀, bearing now a Me₃Si-substituted Cp ligand was used as catalyst, the opposite diastereoisomer was formed as a major product (path b, Scheme 20). Finally, when propargyl amine was treated with **L**₂₀, the combined intermolecular carboamination followed by the intramolecular carbometalation proceeded to give the bicyclic product **25** possessing an exoalkenyl unit as single Z-isomer in outstanding enantioselectivity (Path c, Scheme 20).

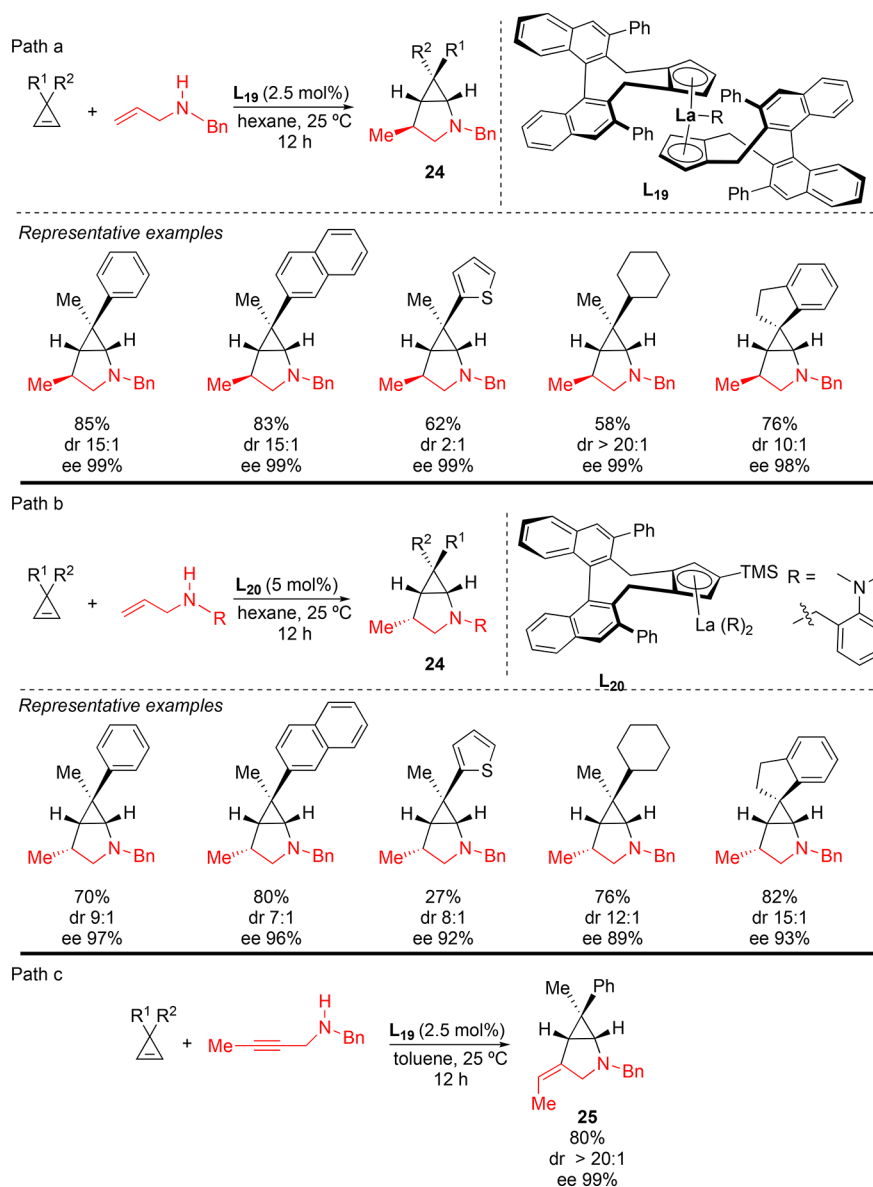
A different approach to prepare cyclopropylamine derivatives consists in the interesting oxime version of the Cope-amination reaction. In this context, the copper-catalyzed retro-cope aminometalation was developed.^{125–130} The (R)-DTBM-Segphos (**L**₅)/copper-catalyzed “hydronitronylation” of cyclo-

propenes proceeded under very mild conditions to give the corresponding optically enriched cyclopropyl nitrones **26** with high enantioselectivities in most cases as single diastereoisomer (Scheme 21). It should be noted that various oximes could be used for this addition reaction with aromatic groups possessing electron rich and electron poor substituents, heteroaryl groups but also aldoximes.¹³¹

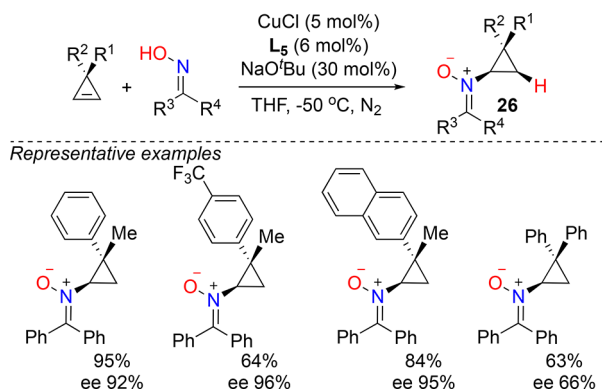
3.7. Direct Functionalization of Achiral Three-Membered Carbocycles by Asymmetric Hydroarylation Reactions

The diastereo- and enantioselective Josiphos/Rh-catalyzed asymmetric arylation reactions of cyclopropene derivatives with commercially available aryl boronic acids provide the arylated cyclopropanes **27** in excellent diastereo- and enantiomeric ratios (Scheme 22).¹³² Aryl boronic acids containing electron-donating or electron-withdrawing groups could be used indifferently as they have nearly no influence on the diastereoselectivity and enantioselectivity of the catalytic reaction. Even aryl boronic acid containing a NO₂ moiety, fluorine substituents, or a trifluoromethyl group gave the desired

Scheme 20. Lanthanum-Catalyzed Diastereodivergent Asymmetric Carboamination—Annulation of Cyclopropenes with Aminoalkenes



Scheme 21. Copper-Catalyzed Cope-Hydroamination of Cyclopropene with Oximes



arylated products with high selectivities. Similarly, cyclopropenes containing various substituents on the aromatic group and even identical substituents undergo the asymmetric arylation reaction with similar selectivities.

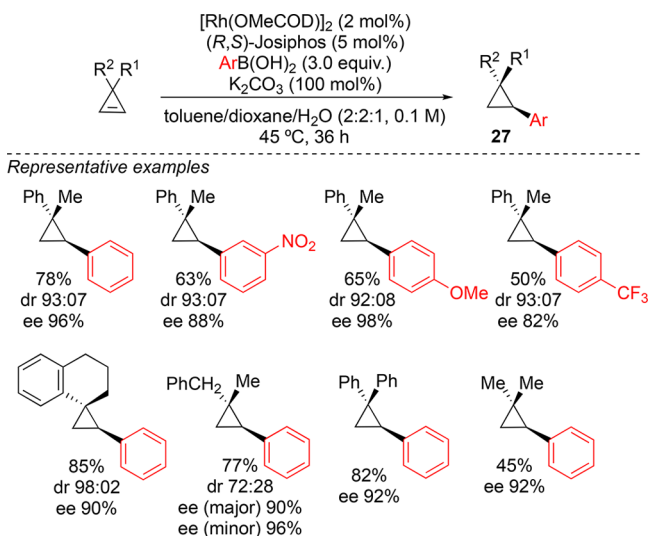
3.8. Direct Functionalization of Achiral Three-Membered Carbocycles by Asymmetric Hydroalkynylation Reactions

Chiral half-sandwich gadolinium complex promotes the first and direct asymmetric hydroalkynylation of achiral cyclopropenes to form the enantioenriched alkynycyclopropanes in good yields and excellent stereoselectivities (Scheme 23).¹³³ A wide range of terminal alkynes containing various substituents could undergo this asymmetric hydroalkynylation reaction affording an interesting chemical handle for subsequent transformations.

3.9. Direct Functionalization of Achiral Three-Membered Carbocycles by Asymmetric Hydroallylation Reactions

An enantioselective protocol for the hydroallylation of 3,3-cyclopropenes employing a ligand derived from Xantphos (**L22**) has been developed utilizing an *in situ* formed copper hydride.

Scheme 22. Rh-Catalyzed Arylation of Cyclopropanes



Two different types of allyl electrophiles could be utilized yielding disubstituted cyclopropanes in decent selectivities (Scheme 24).¹³⁴

3.10. Direct Functionalization of Achiral Three-Membered Carbocycles by Asymmetric Carbometalation Reactions

The carbometalation reaction,^{135–137} as opposed to all reactions described above, consists in the addition of an organometallic species across the double bond to provide a new organometallic species that can be subjected to subsequent synthetic transformations. 1,2-Bisalkylated derivatives are thus obtained. Although carbometalation reactions of alkyne have been extensively developed over the last few decades,¹³⁵ the analogous transformation on alkenes is much more challenging and is usually substrates dependent.^{138,139} As mentioned before, due to the ring strain of the double bond, cyclopropanes behave more like alkynes and carbometalation of cyclopropanes represents an exception for the addition on double bonds. As the resulting cyclopropyl metal species is configurationally

stable, addition of an electrophile should produce the 1,2-disubstituted cyclopropanes.^{140,141} Although the resulting cyclopropylzinc species has never been trapped, one of the first asymmetric $FeCl_3/(R)$ -*p*-Tol-BINAP-catalyzed carbocyclopropanation of cyclopropanes was reported by Nakamura on cyclopropenone ketal in a mixture of solvents toluene/tetrahydropyran (THP) and in the presence of *N,N,N',N'*-tetramethylethylenediamine (TMEDA). It is worth mentioning that the excess amount of diamine ligand is required to obtain high enantioselectivity as racemic products are only produced in its absence (Scheme 25).¹⁴²

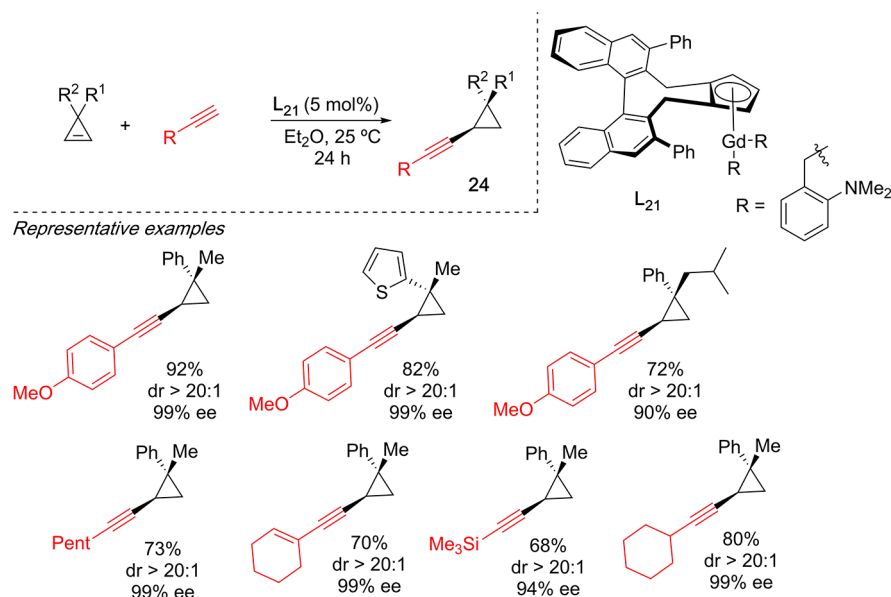
In 2006, Fox and co-workers reported that cyclopropenylcarbinols **30** could undergo a *N*-methylprolinol-mediated methylmagnesium reaction to provide the corresponding cyclopropylcarbinols **31** with high diastereo- and enantioselectivity (Scheme 26). During this study, they found the primordial role of magnesium methoxide to produce consistent and reproducible results. The chiral cyclopropylmagnesium species was subsequently trapped with electrophiles to provide substituted chiral cyclopropanes. However, the enantioselectivity decreased when different alkyl Grignard reagents other than $MeMgCl$ were added to cyclopropenylcarbinols **30** (Scheme 26).¹⁴³

In 2010, Lautens and co-workers reported the first Pd-catalyzed/*(R)*-Tol-BINAP ethylzincation of unfunctionalized spirocyclopropane **32** in high enantioselectivity. The intermediate cyclopropylzinc species could, eventually after transmetalation to copper, be subsequently functionalized with a variety of different electrophiles to deliver the enantioenriched cyclopropyl derivatives **33** (Scheme 27).¹⁴⁴

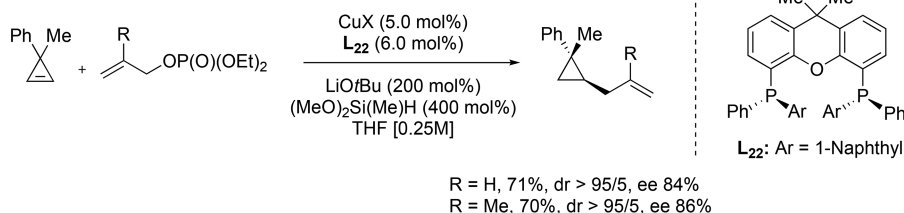
Following this pioneering work, and using the previously described Tortosa conditions,⁴⁶ the asymmetric copper/*(R)*-DTBM-Segphos catalyzed ethylzincation of nonfunctionalized cyclopropanes **17** was developed. The enantioenriched polysubstituted cyclopropanes **34** were obtained with high diastereo- and enantioselectivities (Scheme 28).¹⁴⁵

To extend the scope of this transformation, a large variety of differently substituted cyclopropanes **17** were tested and in all cases, the configurationally stable cyclopropylzinc species were subsequently functionalized to give a large spectrum of ethylated

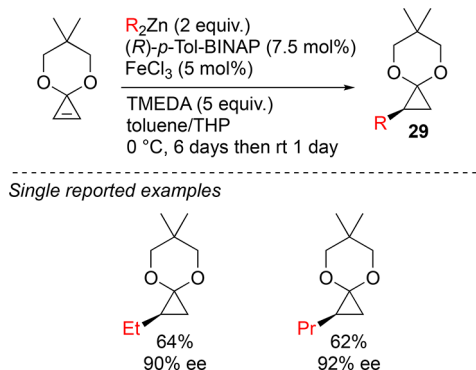
Scheme 23. Gd-Catalyzed Hydroalkynylation of Cyclopropanes with Terminal Alkynes



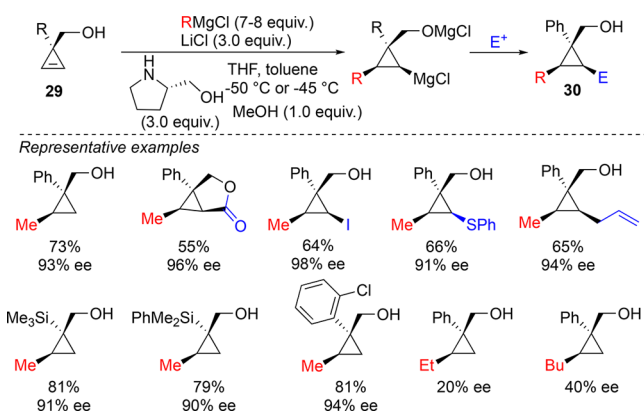
Scheme 24. Enantioselective Copper-Catalyzed Hydroallylation Reaction



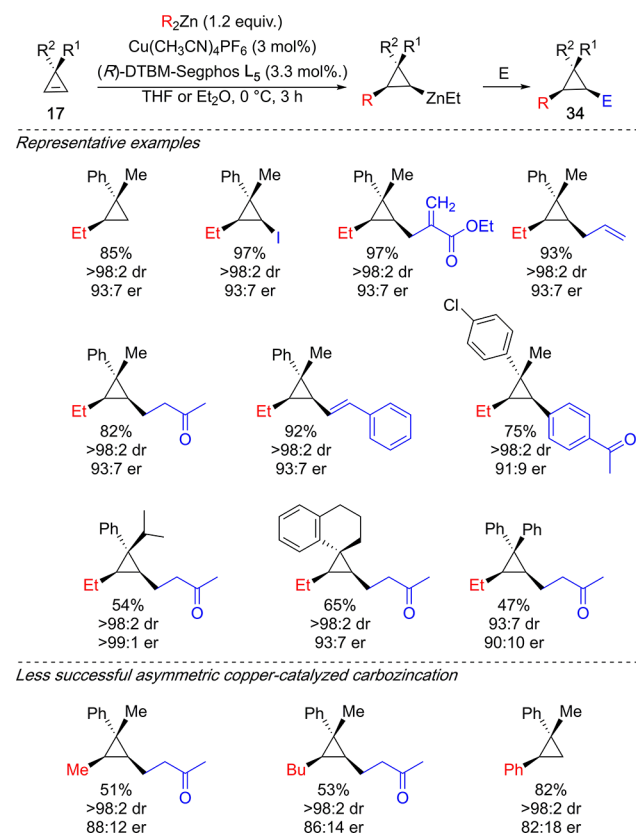
Scheme 25. Iron-Catalyzed Cyclopropene Carbozincation



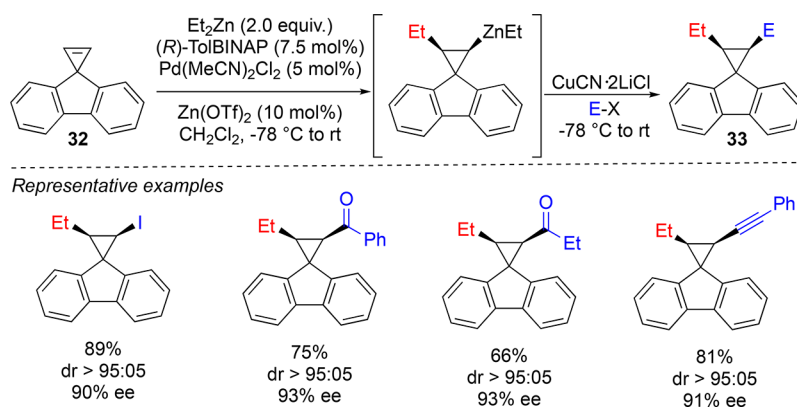
Scheme 26. Enantioselective Carbomagnesiation of Cyclopropenes



Scheme 28. Asymmetric Copper-Catalyzed Carbozincation of Cyclopropenes

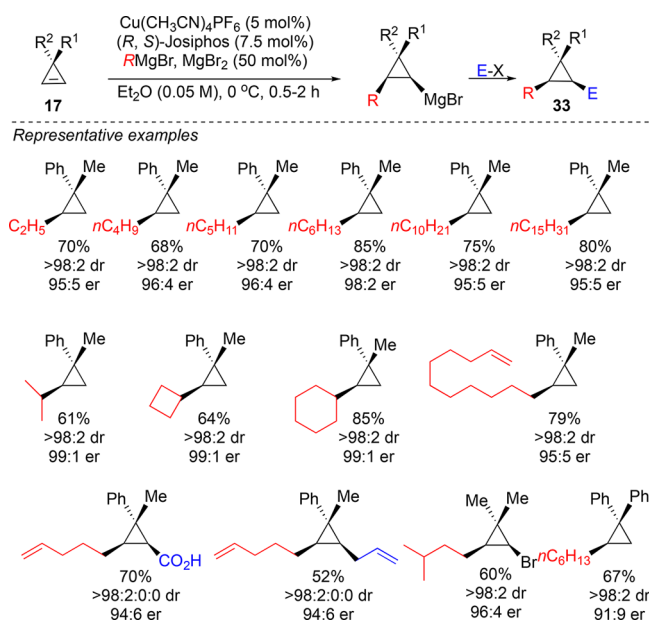


Scheme 27. Enantioselective Palladium-Catalyzed Carbozincation of Cyclopropenes



cyclopropane derivatives. However, when different commercially available dialkylzinc derivatives were used for this asymmetric copper-catalyzed addition, enantiomeric ratios were lower, limiting therefore this transformation to the addition of an ethyl group. To solve this problem, the asymmetric copper/Josiphos-catalyzed carbomagnesiation of cyclopropenes was developed.¹⁴⁶ Although asymmetric catalysis is usually more challenging with Grignard reagents^{147–149} than that with dialkylzinc reagents, a rather large variety of alkyl Grignard

Scheme 29. Asymmetric Copper-Catalyzed Carbomagnesiation of Cyclopropenes



derivatives were added to **17** with outstanding diastereo- and enantioselectivities (Scheme 29).

It is worth mentioning that the addition of MgBr_2 plays an essential role to consistently obtained high enantioselectivities, most probably by shifting the Schlenk equilibrium of the Grignard reagent (the copper-catalyzed addition of R_2Mg led to low enantiomeric ratios).^{150–152} Having in hand an easy protocol for the asymmetric addition of a large variety of

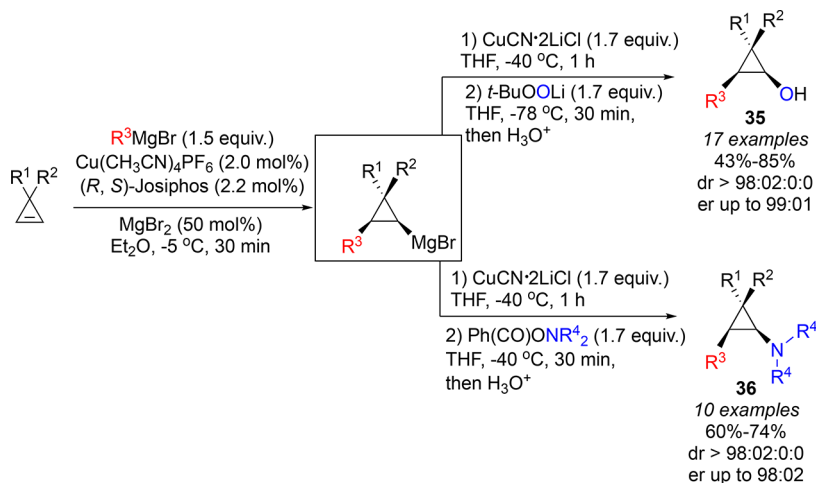
Grignard reagents to cyclopropenes, an electrophilic oxidation or amination reaction of the resulting cyclopropyl Grignard intermediates should then provide corresponding cyclopropanol **35** and cyclopropylamine derivatives **36** with the same diastereo- and enantioselectivities than the carbometalated products, if these last two transformations proceed with a complete retention of configuration.^{153–161} As described in Scheme 30, addition of oxenoids or hydroxylamines to the enantiomerically enriched cyclopropylmagnesium species led to the expected heterosubstituted cyclopropane derivatives **35** and **36**.¹⁶²

Using the same concept of asymmetric copper-catalyzed carbometallation reaction, the vinylmetalation reaction, in the presence of a commercially available chiral ligand $(R)\text{-DTBM-Segphos}$ or alternatively SchmalzPhos as chiral ligand, has been developed providing a direct and easy approach toward the synthesis of chiral vinylcyclopropanes **37** (Scheme 31).¹⁶³ The vinylaluminum species is in situ generated by hydroalumination reaction of a terminal alkyne. It is important to emphasize that a transmetalation reaction of the resulting vinylaluminum species with Et_2Zn is required to avoid degradation of the starting cyclopropene derivatives **17**. Although the diastereoselectivity of this transformation is excellent, the enantioselectivities were only moderate.

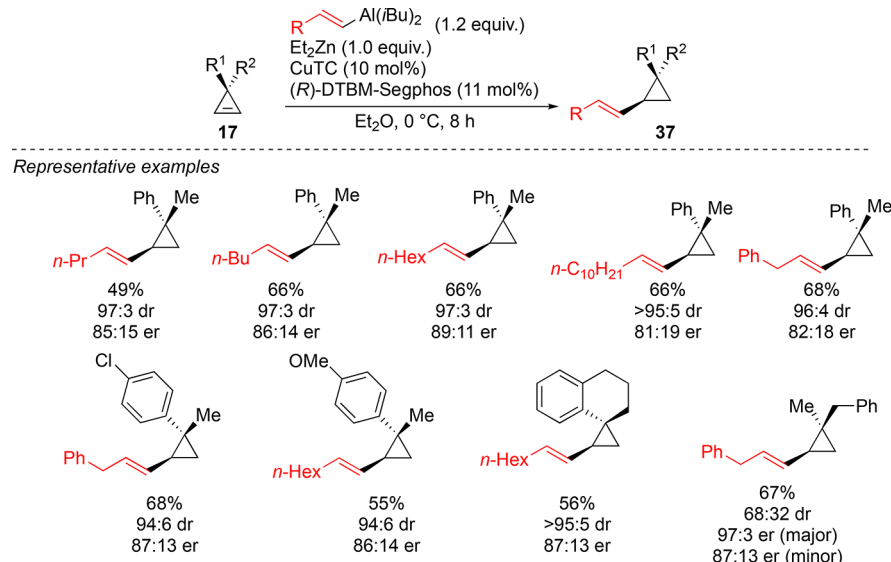
4. CONCLUSION AND OUTLOOK

It became rapidly clear that achiral substrates such as 1,1-disubstituted cyclopropanes or 3,3-disubstituted cyclopropenes could be seen as unique scaffolds for the preparation of a large variety of diastereo- and enantiomerically enriched cyclopropanes. This approach consists in the direct functionalization of three-membered carbocycles either through a C–H bond activation reaction for 1,1-disubstituted cyclopropanes or through selective transformation of the double bond for 3,3-disubstituted cyclopropenes. In all cases, this unique catalytic manipulation of a single precursor allows, at will, the formation of all types of polysubstituted cyclopropanes such as alkylcyclopropanes, vinylcyclopropanes, arylcyclopropanes, alkynylcyclopropanes, cyclopropanols, and cyclopropylamines. We believe that this proposed alternative approach to the asymmetric and catalytic preparation of polysubstituted cyclopropanes would certainly be useful and represent a powerful addition into the research field of small ring chemistry. However,

Scheme 30. Asymmetric Catalytic Preparation of Polysubstituted Cyclopropanol and Cyclopropylamine Derivatives



Scheme 31. Tandem Hydroalumination/Cu-Catalyzed Asymmetric Vinylmetalation of Cyclopropenes



there are still many challenges that need to be addressed before becoming a fully general approach; for instance, no examples have been yet reported for the catalytic asymmetric addition on 1-substituted cyclopropenes. We have no doubt that these approaches will find numerous applications in the field of total synthesis, for late-stage functionalization of complex natural products and pharmaceutical drugs. To our delight, some very elegant applications have been discussed in this Focus Review establishing the proof-of concept and more applications for the construction of complex molecular skeletons from these simplest and smallest carbocycles that could be expected.

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Notes

The authors declare no competing financial interest.

Biographies

Dr. Longyang Dian obtained his B.S. degree in Chemistry from Sun Yat-Sun University (2010) and then received his M.Sc. degree (Medicinal Chemistry) with Prof. Xiaoguang Lei from Tianjin University & National Institute of Biological Science, Beijing (NIBS), in 2012. He then pursued his doctoral studies with Prof. Kang Zhao and Prof. Yunfei Du in Tianjin University working on radical involved oxidative $C(sp^3)-H$ functionalization. After receiving his Ph.D. degree in 2015, Longyang joined the research group of Prof. Ilan Marek at the Technion – Israel Institute of Technology, in Israel. He is currently developing the asymmetric functionalization of three-membered unsaturated carbocycles.

Prof. Ilan Marek, FRSC, was born in Haifa (Israel), educated in France, and received his Ph.D. thesis in 1988 from the University Pierre et Marie Curie, Paris, (France) under the guidance of Professor J. F. Normant. In 1989, he did a short postdoctoral stage in Louvain-la-Neuve (Belgium) with Professor L. Ghosez before obtaining a research position at the CNRS in France in 1990. He then moved to the Technion – Israel Institute of Technology at the end of 1997 where he

currently holds a full Professor position. Since 2005, he holds the Sir Michael and Lady Sobell Academic Chair and has been an elected member of the French Academy of Science since 2017. He serves on many advisory boards of journals including *Chemical Reviews* from the American Chemical Society.

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